CHAPTER 3

CALL ADMISSION CONTROL

When designing communication networks that support services with specific Quality of Service (QoS) requirements, it is important to include admission control mechanisms that regulate the number of information flows allowed into the system. The objective is to prevent congestion and overloading of shared network resources, which may result in the degradation of the Quality of Experience (QoE) experienced by some users below a minimum acceptable threshold.

Such mechanisms are generally referred to as *Call Admission Control (CAC)*. Their role is to determine whether a new Information Flow (generally referred to as a *call* in the CAC terminology) can be admitted into the system without violating the Service Level Agreement (SLA) of any of the active calls, including the one requesting admission.

To make this decision, CAC algorithms require information about the traffic characteristics and the QoS requirements of each call. Clearly, admitting a new call may require a reallocation of network resources. Therefore, the CAC should be co-designed with the Resource Manager (RM), so that admission decisions account for the resources that will actually be assigned to all active flows after the new call has been admitted.

The design of optimal CAC algorithms is thus a rather complex task, which may require accurate information about the traffic generated by the sources and system-wide control over network resources.

Nonetheless, several relatively simple CAC strategies can be defined that, although not necessarily optimal, are effective in many practical scenarios. Depending on the assumptions adopted to characterize the traffic and the desired service guarantees, these algorithms may be more or less conservative.

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Conservative CAC algorithms may deny admission to calls that could actually be accommodated without violating any SLA, thus reducing network utilization. Conversely, more aggressive algorithms may admit additional calls at the expense of weaker QoS guarantees, potentially resulting in the degradation of the QoE experienced by some users.

In the following, we will review some classical CAC algorithms, which offer different trade-offs between QoS guarantees and network utilization. For simplicity, we will consider admission control for a single link of capacity c, equipped with an input buffer of size B, although the same principles can be extended to networks consisting of multiple links.

3.1 Deterministic Bound

According to this policy, a new Information Flow is admitted into the system only if the aggregate traffic generated by all active calls does not exceed the link capacity, even in the *worst-case scenario*, in which all sources transmit simultaneously at their peak rates.

Therefore, this CAC policy requires each source s to declare its peak rate $\lambda_{p,s}$. Denoting by i the call requesting admission, the CAC algorithm will admit it only if

$$\lambda_{p,i} + \sum_{s \in S} \lambda_{p,s} \leq c, \tag{3.1}$$

where S is the set of Information Flows already admitted into the system. If (3.1) does not hold, the new call is *blocked*, i.e., denied access to the system.

Provided that all sources respect their declared peak rates, this condition guarantees that the aggregate instantaneous traffic generation rate never exceeds the link capacity, thereby preventing congestion due to insufficient link bitrate.

Example 3.1 A Consider a link with bitrate $c = 10$ Mbit/s and an input buffer of size $B = 1$ Mbit. Assume that the link is shared by bursty sources following an ON-OFF traffic generation pattern. The ON periods have a constant duration $T_{ON} = 1$ ms, during which each source transmits at the constant peak rate $G_{ON}(t) = \lambda_{p,s} = 1$ Mbit/s.

The OFF periods have exponentially distributed durations, with mean $T_{OFF} = 1$ s, during which the sources are idle, i.e., $G_{OFF}(t) = 0$.

Using the CAC criterion (3.1), the maximum number C_1 of calls that can be admitted into the system is

$$C_1 = \left\lfloor \frac{c}{\lambda_{p,s}} \right\rfloor = 10\,. \tag{3.2}$$

Notice that the average traffic generated by each source is approximately 1 kbit/s. Therefore, the aggregate average traffic generated by the ten admitted sources is only about 10 kbit/s, corresponding to 0.1% of the link capacity.

This example highlights the potentially poor resource utilization resulting from a worst-case admission criterion.

A less restrictive CAC algorithm can be obtained by requiring each source to conform to a traffic envelope defined by a Token Bucket Filter (TBF). In particular, each source s must declare the following parameters:

- Token generation rate: ρ_s;
- Bucket size: b_s.

These parameters define an upper bound on the amount of traffic that the source is allowed to generate over any time interval of duration τ:

$$A_s(t+\tau) - A_s(t) \leq \rho_s \tau + b_s \,,$$

where $A_s(t)$ denotes the cumulative amount of data generated by source s up to time t.

The parameters ρ_s and b_s should be selected according to the traffic characteristics of the source. Traffic exceeding the declared envelope must be delayed by a shaper or handled according to an appropriate traffic policing policy. A possible CAC algorithm is then the following. Denoting by B the size of the link's input buffer, a new call i is admitted into the system if both of the following conditions hold:

1. $c \geq \rho_i + \sum_{s \in S} \rho_s$;
2. $B \geq b_i + \sum_{s \in S} b_s$.

The first condition requires that the link bitrate be at least equal to the aggregate committed information rate of the active sources. The second condition guarantees that the link buffer does not overflow, even in the worst-case traffic scenario, provided that all sources conform to their declared traffic envelopes and the link operates as a work-conserving server with constant service rate c. Indeed, denoting by $\rho_{\rm tot}$ and $b_{\rm tot}$ the sums of the token generation rates and bucket sizes of all active sources, respectively, the aggregate traffic generated over any interval of duration τ is bounded by

$$A(t+\tau) - A(t) \leq \rho_{\rm tot} \tau + b_{\rm tot} \,.$$

The maximum amount of data queued at the link can therefore be bounded as

$$Q_{\max} \leq \sup_{\tau \geq 0} \{\rho_{\rm tot} \tau + b_{\rm tot} - c\tau\} \,.$$

Since the first admission condition guarantees that $\rho_{\rm tot} \leq c$, we obtain

$$Q_{\max} \leq b_{\rm tot} \,.$$

Consequently, the second admission condition provides a deterministic guarantee against buffer overflow. Note, however, that this condition can be very conservative when the sources are statistically independent and their traffic bursts occur asynchronously. In such cases, the probability that the aggregate traffic approaches its worst-case envelope may be extremely small. Therefore, even when the second condition is not satisfied, the probability of buffer overflow may remain negligible. This observation motivates the introduction of *aggressive* CAC schemes, which provide probabilistic QoS guarantees rather than deterministic bounds, as discussed in the following sections.

Example 3.1 B Consider the scenario of the previous example. We can select the TBF parameters according to the average traffic generation rate and the typical burst size of the sources. The token generation rate ρ should be at least equal to the average traffic generated by each source, i.e.,

$$\rho \geq \frac{\bar{G}_{ON} T_{ON} + \bar{G}_{OFF} T_{OFF}}{T_{ON} + T_{OFF}} \simeq 1 \,\text{kbit/s} \,,$$

where the approximation is by excess and is therefore conservative in terms of the committed information rate. The bucket size should be sufficient to accommodate the excess traffic generated during

an ON period. Since each source transmits at rate $\lambda_{p,s} = 1$ Mbit/s during an ON period of duration $T_{ON} = 1$ ms, the amount of tokens required to accommodate an isolated ON period, assuming an initially full bucket, is

$$b_{\min} = (\lambda_{p,s} - \rho)T_{ON} \simeq 1 \text{ kbit}\,.$$

The approximation is again by excess, and we can therefore select $b_s = 10^3$ tokens, assuming that each token corresponds to one bit. Notice, however, that the OFF periods are exponentially distributed and may occasionally be very short. Therefore, consecutive ON periods may occur before the token bucket has been completely replenished. Consequently, the selected bucket size does not guarantee that every realization of the original ON-OFF source process conforms to the declared traffic envelope. Such a guarantee would require an appropriate shaping or policing mechanism. Assuming that the traffic admitted into the link conforms to the declared token-bucket parameters, the maximum number C_2 of calls that can be admitted using the second CAC criterion is

$$C_2 = \min\left\{C_2', C_2''\right\},$$

where

1. $C_2' = \lfloor c/\rho_s \rfloor \simeq 10000\,;$
2. $C_2'' = \lfloor B/b_s \rfloor \simeq 1000\,.$

Therefore, we obtain $C_2 = 1000$ calls, with the limiting factor being the size of the link buffer. By increasing the buffer size, the number of admissible calls could be further increased, up to the limit imposed by the link bitrate. Admitting more than C_2'' calls would violate the deterministic buffer condition, so that buffer overflow could no longer be excluded in the worst case. Nevertheless, if the sources generate statistically independent traffic, the actual probability of buffer overflow may remain very small, even when the number of admitted calls exceeds C_2''. This illustrates the potential advantages of statistical multiplexing, which will be exploited by the probabilistic CAC algorithms introduced in the following sections.

3.2 Statistical bounds

The deterministic CAC methods described in the previous section are simple and provide strict QoS guarantees to the sources admitted into the system. However, they may lead to significant underutilization of network resources, particularly in the presence of bursty traffic sources.

For example, consider a group of N statistically independent and identically distributed ON-OFF sources. The ON and OFF periods have random durations with means T_{ON} and T_{OFF}, respectively. During the ON phase, each source s generates data at a constant bitrate $\lambda_{p,s}$.

Let $n(t)$ denote the number of sources that are in the ON state at time t. Then, we have

$$\mathrm{P}\left[n(t) \geq k\right] = \sum_{j=k}^{N} \binom{N}{j} P_{ON}^{j} (1 - P_{ON})^{N-j} \tag{3.3}$$

where $P_{ON} = T_{ON}/(T_{ON} + T_{OFF})$ is the probability that a source is in the ON state at an arbitrary observation time. Eq. (3.3) is plotted in solid lines in Fig. 3.1 as a function of k, for $N = 1000$ and different values of P_{ON}. We observe that, for $P_{ON} = 0.7$, the probability that at least 750 sources are simultaneously active is less than 10^{-4}.

In particular, the probability that all N sources are simultaneously in the ON state is P_{ON}^N, which decreases exponentially with N for $P_{ON} < 1$.

The average number of active sources is NP_{ON}, which can be significantly smaller than N for highly bursty sources (i.e., $P_{ON} \ll 1$).

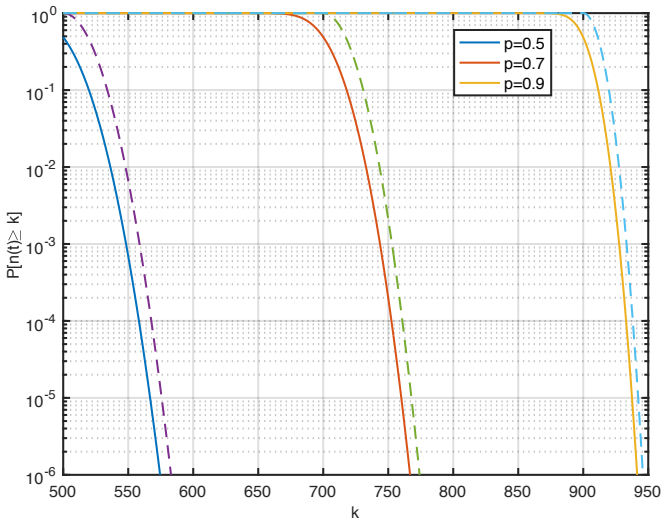


Figure 3.1 Tail of the binomial distribution for $N = 1000$ and different values of p: exact probability (solid lines) and Chernoff upper bound (dashed lines).

To improve network resource utilization, statistical CAC methods may admit more sources than deterministic methods by exploiting *statistical multiplexing*, i.e., the fact that the aggregate traffic rate is typically much lower than the sum of the individual peak rates, particularly when the sources generate bursty traffic. However, when the aggregate traffic rate exceeds the available link capacity, data may accumulate in the buffer, potentially resulting in *outage*, i.e., temporary violations of the QoS constraints, such as excessive delays, packet losses, or reduced throughput.

Statistical CAC strategies are therefore designed to keep the probability of such violations below a prescribed threshold ε.

Since several QoS parameters, particularly packet delay and loss probability, are closely related to buffer occupancy, a common approach to statistical CAC is to constrain the probability that the amount of queued data exceeds a given threshold.

Formlly, assume that $\{X_s(\tau), s \in S\}$ are i.i.d. r.v.s. The aggregate average arrival rate generated by the $N = |S|$ sources is then

$$G(\tau) = \sum_{s=1}^{N} X_s(\tau)\,. \tag{3.4}$$

Consider a link with transmission capacity c and a buffer of size B. During an observation interval of duration τ, the aggregate amount of arriving data is $G(\tau)\tau$, while the link can transmit at most $c\tau$ units of data.

Therefore, the quantity

$$[G(\tau) - c]\tau$$

represents the difference between the amount of data arriving during the observation interval and the service capacity available over the same interval.

Assuming stationary traffic processes and a work-conserving link,[1] the amount of queued data at an arbitrary observation time is given by the maximum of this difference over all preceding observation intervals, i.e.,

$$Q = \sup_{\tau \geq 0} [G(\tau) - c]\tau \,,$$

where all observation intervals are assumed to end at the same arbitrary time.

A possible statistical admission criterion is then to require that the probability of the queue occupancy exceeding the buffer size B remains below a prescribed threshold ε:

$$\Pr\left(\sup_{\tau \geq 0} [G(\tau) - c]\tau > B\right) \leq \varepsilon \,. \tag{3.5}$$

For given values of τ, B, and ε, the minimum capacity c that satisfies Eq. (3.5) can be interpreted as the *equivalent capacity* required to support the aggregate traffic under the specified statistical constraint.

3.2.1 Statistical CAC without link buffer

Consider a group of N statistically independent and identically distributed ON-OFF sources, operating in their stationary regime. The ON and OFF periods have random durations with means T_{ON} and T_{OFF}, respectively. During the ON phase, each source $h = 1, 2, \ldots, N$ generates Constant Bit Rate (CBR) traffic at a constant rate r. During the OFF phase, the traffic rate is zero.

For simplicity, let us first consider a link of capacity c with no buffer, i.e., $B = 0$.

Let $X_h(t) = 1$ if the hth source is in the ON state at time t, and $X_h(t) = 0$ otherwise. Then, $n(t) = \sum_{h=1}^{N} X_h(t)$ denotes the number of sources simultaneously in the ON state at time t. For ease of notation, in the following we omit the explicit dependence on t, since we consider the system state at an arbitrary observation time.

The CAC algorithm admits N sources into the system if

$$\mathrm{P}\left[n > n^0\right] \leq \varepsilon \tag{3.6}$$

where $n^0 = \lfloor c/r \rfloor$ is the maximum number of simultaneously active sources that can be supported by the link. If the number of sources in the ON state exceeds n^0, the aggregate offered traffic exceeds the link capacity. Since no buffering is available, some packets must be discarded, resulting in an outage.

Because the sources are statistically independent and stationary, the r.v.s $\{X_h\}$ are i.i.d. Bernoulli random variables with parameter $p = P_{ON}$. Consequently, n is a binomial r.v. with parameters N and p, whose PMD is given by (3.3). Its Complementary Cumulative Distribution Function (CCDF), for any $a \in \{0, 1, \ldots, N-1\}$, is

$$\mathrm{P}\left[n > a\right] = \sum_{k=a+1}^{N} \binom{N}{k} P_{ON}^{k} (1 - P_{ON})^{N-k} \,, \tag{3.7}$$

[1] A link is work-conserving if it operates at its full transmission capacity whenever there is data waiting to be transmitted.

while it is zero for $a \geq N$. Hence, (3.6) can be written as

$$\sum_{k=n^0+1}^{N} \binom{N}{k} P_{ON}^k (1 - P_{ON})^{N-k} \leq \varepsilon \,. \tag{3.8}$$

Assuming that (3.8) holds for a given number N of admitted sources, upon the arrival of a new call the CAC algorithm checks whether the condition would still hold for $N + 1$ sources. If so, the new call is admitted; otherwise, it is blocked.

Although the binomial CCDF can be evaluated efficiently using numerical methods, its exact expression may be inconvenient for analytical dimensioning, particularly when dealing with a large number of sources. An alternative is to use an upper bound on the CCDF and impose the outage constraint on this bound. This approach yields a more conservative CAC policy, since the maximum number of sources admitted into the system may be smaller than that allowed by the exact probability constraint. The looser the bound, the larger the potential gap between the conservative policy and the optimal one. Such conservativeness may also provide an additional margin against deviations from the assumed traffic model or other system non-idealities.

A simple upper bound on (3.6) can be obtained by applying the Chernoff bound for binomial r.v.s (A.8). Since n is integer-valued, we have

$$\mathrm{P}\left[n > n^0\right] = \mathrm{P}\left[n \geq n^0 + 1\right] \leq \exp\left(-ND\left(\frac{n^0+1}{N}\middle\| P_{ON}\right)\right) \tag{3.9}$$

where

$$D(q\|p) = q \log \frac{q}{p} + (1-q) \log \frac{1-q}{1-p} \tag{3.10}$$

is the binary relative entropy. The bound is valid in the upper-tail regime $(n^0 + 1)/N > P_{ON}$, with $n^0 < N$. Outside this regime, the trivial upper bound of one can be used. If $n^0 \geq N$, the outage probability is zero.

Example 3.2 A

Consider the scenario of the previous examples.

The sources are ON-OFF, with $P_{ON} = T_{ON}/(T_{ON} + T_{OFF}) \simeq 10^{-3}$ and a rate $r = 1$ Mbit/s during the ON phase.

In Fig. 3.2, we report the tail of the probability distribution of $n(t) = \sum_{h=1}^{N} X_h(t)$. The solid lines represent the exact probabilities, while the dashed lines correspond to the Chernoff upper bound. The vertical red line marks the normalized link capacity $n^0 = \lfloor c/r \rfloor = 10$.

For $N = 1000$, the actual outage probability is approximately $8 \cdot 10^{-9}$, while the Chernoff bound yields a more conservative estimate. For $N = 2000$, the actual outage probability increases to approximately $8 \cdot 10^{-6}$. The Chernoff bound provides an upper estimate in both cases, at the expense of some conservativeness.

Let $N^*(n^0, \varepsilon)$ denote the maximum number of sources admitted by the CAC algorithm, assuming a link with normalized capacity $n^0 = \lfloor c/r \rfloor$ and a target outage probability ε. Fig. 3.3 shows $N^*(n^0, \varepsilon)$ for $n^0 = 10$ and different values of ε. The solid lines correspond to the exact outage probability, while the dashed lines are obtained using the Chernoff upper bound. For a target outage probability of $\varepsilon = 10^{-4}$, the Chernoff-based admission criterion allows fewer sources than the exact criterion, as expected. Nevertheless, both statistical approaches can admit significantly more sources than deterministic CAC, which would limit the number of admitted sources to $n^0 = 10$.

The price paid for this increased admission capacity is that the admitted sources have a non-zero probability of experiencing outage periods, during which the aggregate offered traffic exceeds the link capacity and some packets must be discarded.

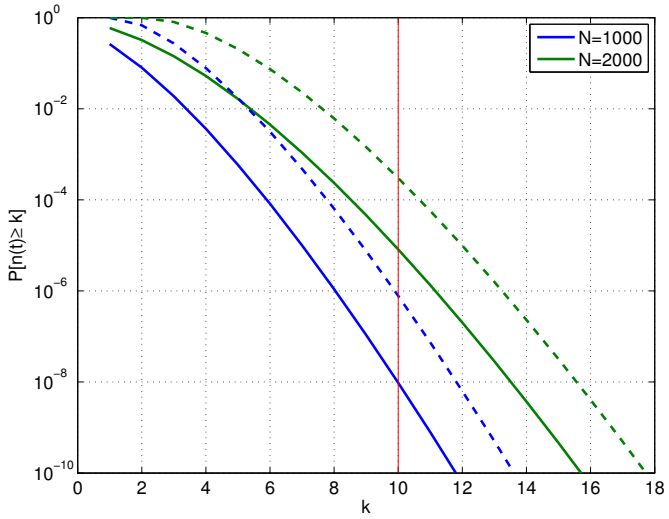


Figure 3.2 Tail of the binomial distribution for $p = 10^{-3}$ and different values of N: exact probability (solid lines) and Chernoff upper bound (dashed lines).

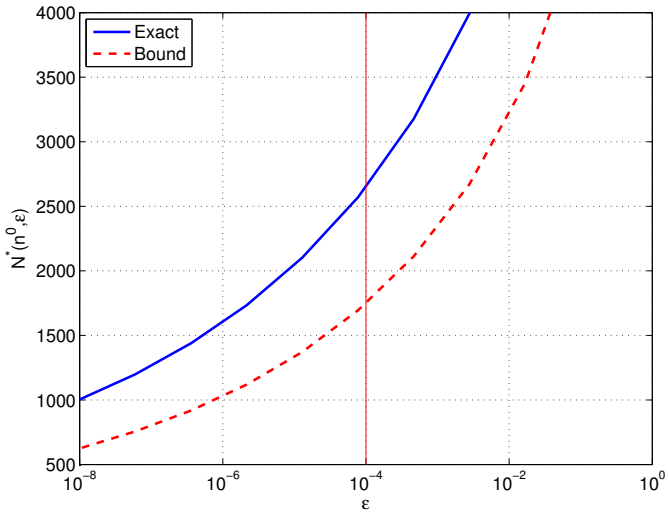


Figure 3.3 Maximum number of admitted calls for a link with normalized capacity $n^0 = 10$, as a function of the outage threshold ε: exact probability (solid lines) and Chernoff-based admission criterion (dashed lines).

3.2.2 Statistical CAC with link buffer

We now consider the case in which the link is equipped with a data buffer that can absorb short-lived traffic bursts, thus making it possible to accommodate a larger number of Information Flows into the system.

For simplicity, we assume that time is slotted, with slots of fixed duration τ. Moreover, all packets are assumed to have equal length, and the link rate c is expressed in packets/s. In each slot, the link can transmit up to $C = \lfloor c\tau \rfloor$ packets.

Let $a_s(t)$ be a r.v. that models the number of packets generated by the s-th source during time slot t. We assume these r.v.s to be i.i.d. across sources and time slots, with common PMD $p_h = \mathrm{P}\left[a_s(t) = h\right]$, for $h \in \mathbb{N}$, and finite mean $\mathrm{m}_{a_s} < \infty$.

Packets generated during a slot are queued and transmitted in First-In First-Out (FIFO) order. We assume a work-conserving link that can serve up to C packets per slot, including packets generated during that slot. Any packets that cannot be transmitted are retained in the queue for transmission in subsequent slots.

Let $q(t)$ denote the number of queued packets at the beginning of time slot t, i.e., the *queue length* at time slot t. The outage constraint is expressed as

$$\mathrm{P}\left[q(t) > B\right] \leq \varepsilon \,, \tag{3.11}$$

where B is a given queue-length threshold that should be exceeded in no more than a fraction ε of the slots, in the stationary regime. Such a constraint can be used, for example, to limit the probability of excessive queueing delays.

In principle, the CAC algorithm should admit a new Information Flow into the system only if condition (3.11) is satisfied. Unfortunately, the exact computation of the queue-length distribution can be difficult, particularly for traffic sources exhibiting temporal correlations, such as ON-OFF sources.

Alternatively, it is possible to derive an upper bound on the CCDF of $q(t)$, thus obtaining a conservative CAC criterion. Moreover, this approach provides useful insights into the behavior of the outage probability as the buffer threshold B varies.

In the following, we derive one such bound under the simplifying assumption of i.i.d. packet arrivals across sources and time slots.

Let N be the number of Information Flows admitted into the system. The total number of packets generated during slot t is the r.v.

$$a(t) = \sum_{s=1}^{N} a_s(t) \,, \tag{3.12}$$

with mean $\mathrm{m}_a = \mathrm{E}\left[a(t)\right] = N\mathrm{m}_{a_s}$, which is constant over time.

To begin with, we observe that a necessary condition for the stability of the queue is that the mean offered traffic does not exceed the service capacity. In particular, for nondegenerate random arrivals, we require

$$N\mathrm{m}_{a_s} < C \,.$$

Otherwise, if the mean offered traffic exceeds C, the queue length progressively grows without bound, and the probability of exceeding any finite threshold B tends to one. In the following, we hence assume $N < C/\mathrm{m}_{a_s}$, i.e., $\mathrm{m}_a < C$.

Suppose the queue is empty at time slot zero, i.e., $q(0) = 0$. Then, for any $t \geq 0$, the queue-length evolution can be recursively described as

$$q(t+1) = \max\{0, q(t) + a(t) - C\}. \tag{3.13}$$

In turn, $q(t)$ can be expressed in a similar manner, so that

$$q(t+1) = \max\{0, \max\{0, q(t-1) + a(t-1) - C\} + a(t) - C\}. \tag{3.14}$$

This can be written as[2]

$$q(t+1) = \max\{0, a(t) - C, q(t-1) + a(t) + a(t-1) - 2C\}. \tag{3.15}$$

Unfolding the recursion, we obtain

$$\begin{aligned} q(t) &= \max\left\{0, a(t-1) - C, a(t-1) + a(t-2) - 2C, \ldots, \sum_{k=0}^{t-1} a(t-k-1) - tC\right\} \\ &= \max\left\{0, \max_{1 \leq m \leq t}\left\{\sum_{k=0}^{m-1} a(t-k-1) - mC\right\}\right\}. \end{aligned} \tag{3.16}$$

The CCDF of $q(t)$ can then be bounded as

$$\begin{aligned} \mathrm{P}[q(t) > B] &= \mathrm{P}\left[\bigcup_{m=1}^{t}\left\{\sum_{k=0}^{m-1} a(t-k-1) > B + mC\right\}\right] \\ &\leq \sum_{m=1}^{t} \mathrm{P}\left[\sum_{k=0}^{m-1} a(t-k-1) > B + mC\right] \\ &\leq \sum_{m=1}^{t} \frac{\mathrm{E}\left[\exp\left(\theta \sum_{\ell=0}^{m-1} a(\ell)\right)\right]}{\exp(\theta(B+mC))} \\ &= \sum_{m=1}^{t} M_{a_s}(\theta)^{Nm} e^{-\theta(B+mC)} \\ &= e^{-\theta B} \sum_{m=1}^{t} \left(M_{a_s}(\theta)^N e^{-\theta C}\right)^m. \end{aligned} \tag{3.17}$$

Here $\theta > 0$ is an arbitrary parameter for which the moment-generating function $M_{a_s}(\theta) = \mathrm{E}\left[e^{\theta a_s}\right]$ is finite. We assume that such values of θ exist. The first inequality follows from the union bound, the second from the Chernoff bound, while the subsequent equality is obtained by exploiting the independence of the arrivals across sources and time slots.

Now, setting $\Lambda(\theta) \stackrel{\mathrm{def}}{=} \log M_{a_s}(\theta)$, we obtain

$$\begin{aligned} \mathrm{P}[q(t) > B] &\leq e^{-\theta B} \sum_{m=1}^{t} \left(e^{-\theta C + N\Lambda(\theta)}\right)^m \\ &\leq e^{-\theta B} \sum_{m=1}^{\infty} \left(e^{-\theta C + N\Lambda(\theta)}\right)^m. \end{aligned} \tag{3.18}$$

[2]For all $X, Y, W, Z \in \mathbb{R}$, it holds that $\max\{X, Y + \max\{W, Z\}\} = \max\{X, Y + W, Y + Z\}$.

The geometric sum in (3.18) converges if

$$N\Lambda(\theta) - \theta C < 0\,,$$

or, equivalently,

$$\frac{\Lambda(\theta)}{\theta} < \frac{C}{N}\,. \tag{3.19}$$

Eq. (3.19) is therefore a feasibility condition for obtaining a finite bound from (3.18). The quantity

$$\alpha(\theta) = \frac{\Lambda(\theta)}{\theta}$$

is the so-called *effective bandwidth* (or *equivalent bandwidth*) of a source, expressed in packets per slot. It represents the amount of service capacity that must be allocated to each source, according to the statistical characterization associated with the parameter θ.

Therefore, condition (3.19) requires that the aggregate effective bandwidth of the N admitted sources be smaller than the available link capacity C.

If condition (3.19) is satisfied, the geometric series in (3.18) converges, yielding

$$\begin{aligned} \mathrm{P}\left[q(t) > B\right] &\leq e^{-\theta B}\left(\frac{1}{1 - e^{-\theta C + N\Lambda(\theta)}} - 1\right) \\ &= e^{-\theta B}\frac{e^{-\theta C + N\Lambda(\theta)}}{1 - e^{-\theta C + N\Lambda(\theta)}}\,. \end{aligned} \tag{3.20}$$

The right-hand side of (3.20) depends on the buffer threshold B, the number N of flows admitted into the system, and the parameter θ. For any feasible $\theta > 0$, the bound decreases exponentially with B. This shows that increasing the buffer threshold can significantly reduce the probability of excessive queue occupancy.

For a given B, however, the dependence on θ is more involved. Increasing θ makes the factor $e^{-\theta B}$ smaller, but also increases the effective bandwidth $\alpha(\theta)$, which is a non-decreasing function of θ. Consequently, the feasibility condition (3.19) becomes more restrictive as θ increases, and the complete bound in (3.20) is not necessarily monotonically decreasing with θ. The tightest bound of this form can be obtained by minimizing the right-hand side of (3.20) over all feasible values of $\theta > 0$.

To better understand the behavior of the effective bandwidth, suppose now that the r.v. $a_s(t)$ takes values in a finite set $A = \{\alpha_0, \alpha_1, \ldots, \alpha_m\}$, with $0 \leq \alpha_0 < \alpha_1 < \ldots < \alpha_m$, where α_m is the maximum possible number of packets generated by a source in one slot, with probability $p_m > 0$.

For large values of θ, the moment-generating function is dominated by the term corresponding to the maximum arrival rate. Therefore,

$$\begin{aligned} \lim_{\theta\to\infty}\frac{\Lambda(\theta)}{\theta} &= \lim_{\theta\to\infty}\frac{\log M_{a_s}(\theta)}{\theta} \\ &= \lim_{\theta\to\infty}\frac{\log\left(\sum_{h=0}^{m} p_h e^{\theta\alpha_h}\right)}{\theta} \\ &= \lim_{\theta\to\infty}\frac{\theta\alpha_m + \log p_m + o(1)}{\theta} \\ &= \alpha_m\,. \end{aligned} \tag{3.21}$$

Hence, the effective bandwidth approaches the peak number of packets that a source can generate in one slot.

If the number of admitted sources satisfies

$$N\alpha_m \leq C\,,$$

the aggregate arrivals can never exceed the service capacity of the link. Consequently, starting from an empty queue, no packets accumulate, and the outage probability is zero. This corresponds to the deterministic admission criterion, which guarantees that the link can accommodate the aggregate peak traffic without requiring any buffering.

However, such a stringent condition is not necessary to achieve arbitrarily small outage probabilities when buffering is available. Indeed, for a stable queue, the probability of exceeding a given threshold can be made arbitrarily small by choosing a sufficiently large value of B.

Conversely, letting $\theta \to 0^+$, the effective bandwidth approaches the mean number of packets generated by a source in one slot:

$$\begin{aligned}\lim_{\theta\to 0^+}\frac{\Lambda(\theta)}{\theta} &= \lim_{\theta\to 0^+}\frac{\log M_{a_s}(\theta)}{\theta}\\ &= \frac{M'_{a_s}(0)}{M_{a_s}(0)} = \mathrm{m}_{a_s}\,,\end{aligned} \tag{3.22}$$

where $M'_{a_s}(0)$ denotes the derivative of the moment-generating function at zero, which is equal to the first-order moment of the r.v. a_s. Thus, for small values of θ, the effective bandwidth approaches the average source traffic, and condition (3.19) approaches

$$N < \frac{C}{\mathrm{m}_{a_s}}\,.$$

This is the stability condition for the considered queueing system with nondegenerate random arrivals. It determines the maximum number of sources that can be admitted while maintaining a stable queue, although the actual outage probability also depends on the buffer threshold B and on the statistical properties of the traffic.

In particular, when $N\mathrm{m}_{a_s} > C$, the mean offered traffic exceeds the service capacity, and the queue length grows without bound in the long term. Consequently, the probability of exceeding any finite queue-length threshold tends to one.

At the limiting condition $N\mathrm{m}_{a_s} = C$, the queue is generally not positive recurrent for nondegenerate random arrivals, so that a strict inequality is required to guarantee stability.

3.2.3 An application: dimensioning a video play-out buffer

The statistical bounds introduced in the previous sections can be applied to several network dimensioning problems. As an example, we consider the dimensioning of the play-out buffer of a video streaming application, with the objective of limiting the probability of playback interruptions while keeping the initial buffering delay reasonably small.

Example 3.2 B

Consider a video streaming system in which the RM algorithms guarantee that the aggregate bitrate allocated to the active calls does not exceed the available channel capacity. In particular, we assume that the rate allocated to each video stream is sufficient to sustain its average encoding bitrate.

However, the actual amount of data generated by a video stream may fluctuate around its average value, because successive Group of Picturess (GoPs) generally have different sizes. These fluctuations can be absorbed by the play-out buffer at the client side, provided that a sufficient amount of video content is downloaded before starting playback.

The problem is then to determine the minimum amount of initially buffered video content that makes the probability of playback interruptions sufficiently small.

Traffic model.

Let τ_v be the duration of each GoP in video sequence v, and let $s_v^h(\kappa)$ denote the size of the hth GoP when the video is encoded at compression level κ.

In principle, the sizes of the different GoPs can be determined by the video server and communicated to the client through the Media Presentation Description (MPD). For simplicity, however, we model the sizes of successive GoPs as independent and identically distributed r.v.s, with mean $s_v(\kappa) = \mathrm{E}\left[s_v^h(\kappa)\right]$ and standard deviation $\sigma_v(\kappa)$.

Furthermore, we assume that the GoP sizes are bounded, and denote by $\Delta_v(\kappa)$ the difference between their maximum and minimum possible values. The parameters required by the statistical analysis are assumed to be available to the client.

Let $\gamma_v^\circ R$ denote the transmission rate allocated to video stream v, which is assumed to remain constant during playback.

We further assume that the RM algorithm selects the compression level κ so that the average encoding bitrate is equal to a fraction $\alpha \in (0, 1]$ of the allocated transmission rate, i.e.,

$$\kappa \;:\; \frac{s_v(\kappa)}{\tau_v} = \alpha\gamma_v^\circ R\,.$$

The parameter α therefore represents the fraction of the allocated transmission capacity used, on average, by the video stream. For $\alpha < 1$, the unused fraction provides a margin that can compensate for temporary increases in the encoding bitrate.

Playback model and outage probability.

Let n_0 be the number of GoPs buffered by the client before starting playback.[3] Once playback starts, one GoP is consumed every τ_v seconds, while new GoPs are downloaded from the network.

Consider the download of n new GoPs after playback has started. Their aggregate size is the r.v.

$$S_v(n;\kappa) = \sum_{h=1}^{n} s_v^h(\kappa)\,,$$

with mean

$$\mu = \mathrm{E}\left[S_v(n;\kappa)\right] = ns_v(\kappa) = n\alpha\tau_v\gamma_v^\circ R\,.$$

[3]As customary, the size of the play-out buffer is here expressed in terms of playing time of the buffered video content. Its actual size in bytes depends on the compression level of the video sequence.

The amount of data that can be downloaded during the playback time of $n_0 + n$ GoPs is

$$D_v(n) = \tau_v \gamma_v^\circ R(n_0 + n)\,.$$

A playback interruption may occur if the aggregate size of the first n downloaded GoPs exceeds the amount of data that can be received before their playback deadline.

We therefore consider the probability

$$P_f(n;\kappa) = \Pr[S_v(n;\kappa) \geq D_v(n)] = \Pr[S_v(n;\kappa) \geq \mu(1+\delta)]\,,$$

where

$$\delta = \frac{n + n_0}{n\alpha} - 1\,.$$

This probability characterizes a potential playback interruption at a given playback horizon n. It does not directly represent the probability of experiencing at least one freezing event during the entire video session, which would require considering the possible violation of all playback deadlines.

Our objective is to determine a value of n_0 such that

$$P_f(n;\kappa) \leq P_f^*$$

for every $n \in \{1, 2, \ldots, n_{\max}\}$, where P_f^* is the maximum acceptable probability of violating the playback deadline at any given horizon, and $n_{\max}$ is the maximum number of GoPs considered in the analysis.

Application of statistical bounds.

The exact probability distribution of $S_v(n;\kappa)$ depends on the distribution of the individual GoP sizes, which may be difficult to determine in practice.

However, since the GoP sizes are assumed to be independent and bounded, we can apply Hoeffding's inequality[4] to obtain

$$P_f(n;\kappa) \leq \exp\left(-\frac{2\delta^2\mu^2}{n\Delta_v(\kappa)^2}\right)\,. \tag{3.23}$$

This bound makes it possible to dimension the play-out buffer without explicitly computing the probability distribution of the aggregate GoP sizes.

Requiring the right-hand side of (3.23) to be no greater than P_f^*, we obtain

$$\begin{aligned} n_0 \geq f_0(n;\alpha) &= \frac{\alpha\Delta_v(\kappa)}{s_v(\kappa)}\sqrt{n\log\left(\frac{1}{\sqrt{P_f^*}}\right)} - n(1-\alpha) \\ &= \beta\sqrt{n} - (1-\alpha)n\,, \end{aligned} \tag{3.24}$$

where, for ease of notation, we have defined

$$\beta = \frac{\Delta_v(\kappa)}{\tau_v\gamma_v^\circ R}\sqrt{\log\left(\frac{1}{\sqrt{P_f^*}}\right)}\,. \tag{3.25}$$

To satisfy the constraint for all playback horizons, we need to choose n_0 such that

$$n_0 \geq \max_{1\leq n\leq n_{\max}} f_0(n;\alpha)\,.$$

[4]Hoeffding's inequality follows from the Chernoff bound by applying Hoeffding's lemma to upper-bound the moment-generating function of bounded random variables and optimizing over the Chernoff parameter.

For $0 < \alpha < 1$, considering n as a continuous variable, the right-hand side of (3.24) reaches its maximum at

$$n^* = \frac{\beta^2}{4(1-\alpha)^2} ,$$

for which

$$f_0(n^*; \alpha) = \frac{\beta^2}{4(1-\alpha)} .$$

If $\alpha = 1$, instead, $f_0(n; \alpha)$ is monotonically increasing with n, and its maximum is attained at $n = n_{\max}$.

Therefore, defining

$$\hat{n} = \begin{cases} \min\{n_{\max}, n^*\}, & 0 < \alpha < 1, \\ n_{\max}, & \alpha = 1, \end{cases}$$

we obtain the following conservative approximation for the required number of initially buffered GoPs:

$$n_0 = \left\lceil \max\left\{0, \beta\sqrt{\hat{n}} - (1-\alpha)\hat{n}\right\}\right\rceil . \tag{3.26}$$

Discussion of the results.

The result shows how the required play-out buffer depends on the statistical variability of the video traffic, the allocated transmission rate, and the target probability of violating a playback deadline.

In particular, for a fixed compression level κ, decreasing α corresponds to increasing the transmission capacity available relative to the average encoding bitrate, thus reducing the amount of initial buffering required.

However, if the available transmission capacity is fixed, reducing α also requires selecting a more compressed version of the video stream.

Consequently, there is a tradeoff between the initial buffering delay and the quality of the streamed video.

The playback duration of the initially buffered content is $n_0\tau_v$. The actual time required to download this content depends on its size and on the allocated transmission rate. Under the assumed constant-rate model, its expected value is

$$\mathrm{E}[T_{\mathrm{start}}] = \frac{n_0 s_v(\kappa)}{\gamma_v^\circ R} = \alpha n_0 \tau_v .$$

Numerical example.

Considering the test videos used in this study, we found that $\Delta_v(\kappa)/s_v(\kappa) \leq 0.35$ for all videos and compression levels.

Setting $\alpha = 1$, which allows the highest video quality for the given transmission capacity, and considering video sequences of up to $n_{\max} = 500$ GoPs, (3.26) yields approximately $n_0 = 10$ GoPs for $P_f^* = 5\%$, and $n_0 = 12$ GoPs for $P_f^* = 1\%$.

For GoPs consisting of 12 frames, with a duration of approximately 0.36 seconds, these values correspond to about 3.6 and 4.3 seconds of buffered video content, respectively.

The example illustrates how statistical bounds can be used to derive conservative dimensioning criteria from a limited characterization of the traffic sources, without requiring the exact probability distribution of the aggregate traffic.